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Motion-Aware Personalized ECG Anomaly Detection for False-Alarm Reduction in Wearable Cardiac Risk Screening

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ABSTRACT Continuous ambulatory cardiac monitoring using wearable devices has emerged as a cornerstone for decentralized preventive cardiology, enabling early interception of life-threatening arrhythmias, paroxysmal events, and occult cardiovascular deterioration. Nonetheless, practical adoption in free-living environments remains severely impeded by catastrophic false-alarm rates. Wearable biosensors are chronically susceptible to motion-induced electrode contact disturbances, skin-stretch potentials, electromyographic contamination, and photoplethysmographic (PPG) optical coupling perturbations during activities of daily living. Furthermore, pronounced inter-subject variability in baseline cardiac electrophysiology causes conventional static population-wide machine learning classifiers to flag benign individual-specific variants as pathological crises. Frequent spurious notifications induce debilitating clinical alarm fatigue, prompt user non-compliance, and overwhelm tele-health infrastructures. To resolve these challenges, this paper presents an end-to-end motion-aware personalized electrocardiogram (ECG) anomaly detection framework. The proposed system synchronously ingests Lead-I/II ECG, optical PPG, and triaxial accelerometer telemetry. A multidimensional Signal Quality Index (SQI) module is combined with accelerometer-derived physical activity state estimation to construct a dual-gating decision mechanism. This mechanism dynamically partitions incoming telemetry into accepted, inconclusive, and motion-corrupted regimes. High-quality segments are fed to an activity-conditioned personalized cardiac baseline that adapts using bounded exponentially weighted moving averages across resting and exertional states. By updating only on quality-certified observations, the architecture prevents motion artifacts from corrupting the patient-specific normal reference. A temporal confirmation engine enforces multi-beat persistence and cross-modal hemodynamic verification before escalating to a clinical alert. Validated against extensive clinical and ambulatory cohorts—including CACHET-CADB, WESAD, BUT QDB, PTB-XL, and the MIT-BIH Arrhythmia Database—the proposed system achieves an unprecedented 91.6% reduction in false alarms relative to static population models and a 68.4% reduction compared to unconstrained personalized models, while sustaining 97.2% diagnostic sensitivity, 96.8% specificity, and an ultra-low inference latency of 12.4 ms on wearable ARM Cortex-A53 and Cortex-M4 edge platforms.

INDEX TERMS Wearable cardiac screening, electrocardiogram (ECG), photoplethysmogram (PPG), motion artifact mitigation, personalized baseline modeling, signal quality index (SQI), false-alarm reduction, embedded edge computing.

I. INTRODUCTION

Cardiovascular diseases (CVDs) remain the preeminent cause of global mortality, accounting for an estimated 17.9 million deaths annually according to the World Health Organization [11]. Among these, sudden cardiac arrest (SCA), fatal ventricular arrhythmias, and paroxysmal atrial fibrillation (AFib) represent acute manifestations that frequently strike without prior symptomatic warning. Traditional clinical diagnostics rely primarily on 12-lead resting electrocardiograms (ECG) obtained during clinical visits or 24- to 48-hour Holter monitoring sessions. However, these intermittent modalities exhibit profound diagnostic blind spots: episodic arrhythmias, transient ischemic episodes, and exertion-triggered conduction blocks are statistically improbable to manifest during

brief monitoring intervals, resulting in delayed interventions and adverse patient outcomes.

The rapid miniaturization of low-power microelectronics, flexible dry-contact electrodes, optical reflectance biosensors, and low-energy wireless transceivers has catalyzed a revolution in consumer-grade and clinical wearable telemetry. Devices encompassing smartwatches, adhesive chest patches, smart garments, and ring form-factors now provide the technological substrate for continuous, longitudinal cardiopulmonary surveillance over weeks and months. This continuous sensing paradigm provides an unprecedented window into ambulatory electrophysiology, capturing dynamic fluctuations influenced by circadian cycles, emotional stressors, and physical exertion.

A. THE FALSE-ALARM CRISIS AND ALARM FATIGUE

Despite the undeniable diagnostic potential of continuous wearable cardiac screening, real-world translation from laboratory prototypes to reliable clinical instruments has encountered a profound barrier: the false-alarm crisis. In uncontrolled free-living environments, ambulatory users perform an unconstrained spectrum of physical activities, including ambulation, stair climbing, manual labor, running, and postural reorientation. These movements generate severe motion artifacts (MA) in biological recordings through several physical mechanisms:

1) **Electrode-Skin Impedance Modulation:** Dynamic friction, mechanical deformation, and skin stretching alter the half-cell potential between epidermal layers and metallic or conductive polymer electrodes, generating high-amplitude baseline wander (0.1–1.5 Hz) and transient spikes that mimic premature ventricular contractions (PVCs).
2) **Electromyographic (EMG) Interference:** Muscular contractions in the pectoralis major and upper limbs radiate high-frequency bioelectrical potentials (20–150 Hz) that corrupt the high-frequency components of the QRS complex, compromising morphological feature extraction.
3) **Optical Hemodynamic Decoupling:** In optical photoplethysmography (PPG), acceleration of the sensor chassis relative to capillary beds causes erratic microvascular tissue displacement and ambient light leakage, destroying pulsatile waveform fidelity.

When these corrupted biosignals are processed by standard population-based machine learning models, spurious morphological distortions are frequently misclassified as lethal arrhythmias, yielding false positive rates exceeding 2–5 alarms per monitoring hour. Over a 24-hour monitoring epoch, a patient may be subjected to dozens of ungrounded emergency notifications. This inundation precipitates severe clinical alarm fatigue: patients experience heightened anxiety, sleep disruption, and cognitive exhaustion, ultimately prompting over 40% of users to discontinue device usage within the first month of prescription. Simultaneously, remote cardiac telemetry centers and primary care clinicians are overwhelmed by millions of non-actionable notifications, diluting vigilance and jeopardizing rapid response to authentic medical emergencies.

B. INTER-SUBJECT ELECTROPHYSIOLOGICAL HETEROGENEITY

The false-alarm crisis is exacerbated by profound inter-subject variability in human cardiac electrophysiology. Morphological landmarks on an electrocardiogram are governed by ventricular geometry, myocardial thickness, chest cavity anatomy, autonomic tone, age, biological sex, electrolyte balances, and pre-existing subclinical conduction abnormalities. Consequently, a QRS duration or ST-segment level that signifies imminent ischemia or bundle branch block in one subject may represent a completely benign, stable physiological variant in another.

Conventional detection pipelines employ rigid population-average classifiers trained on large public databases (e.g., MIT-BIH or PTB-XL). These static classifiers impose fixed multi-dimensional decision boundaries. When an individual wearer possesses an atypical native QRS morphology or high vagal tone, a population classifier consistently misinterprets their normal rhythm as anomalous, generating a ceaseless cascade of false alerts even in the complete absence of physical motion. Conversely, if decision thresholds are relaxed to suppress false positives, clinical sensitivity to subtle, genuine paroxysmal arrhythmias collapses to unacceptable levels.

C. PROPOSED SOLUTION AND CONTRIBUTIONS

To overcome these limitations, this paper establishes a novel paradigm: *Motion-Aware Personalized ECG Anomaly Detection*. The foundational thesis of this work is that false-alarm reduction cannot be resolved solely through deeper neural network architectures operating on isolated ECG waveforms. Instead, robust screening requires an interconnected multi-sensor architecture where inertial physical context and multi-channel signal quality indices jointly govern both real-time anomaly inference and continuous personal-baseline adaptation.

The core innovations and contributions of this study are as follows:

- **Multimodal Quality and Motion Fusion Pipeline:** We design a comprehensive preprocessing and feature-extraction engine that synchronously processes Lead-I/II ECG, optical PPG, and triaxial accelerometer telemetry. A battery of complementary statistical, spectral, and morphology-consistency metrics are fused into a composite Signal Quality Index (SQI) alongside physical activity intensity.
- **Dual-Gating Decision Logic:** We formulate an operational three-zone triage policy (Accepted, Inconclusive, Rejected) that dynamically prevents motion-corrupted signals from contaminating the anomaly detector and selectively defers ambiguous windows during heavy exertion without triggering spurious alerts.

- **Activity-Conditioned Personalized Baseline Architecture:** We construct a personalized cardiac electromechanical reference initialized from population priors and updated via bounded Exponentially Weighted Moving Averages (EWMA). By maintaining segregated baseline branches for resting and active states, the framework tracks patient-specific physiological drift while mathematically guaranteeing immunity against catastrophic baseline corruption.
- **Temporal Confirmation Hysteresis Engine:** We formulate a multi-beat persistence and cross-modal hemodynamic validation module that suppresses isolated transient glitches while ensuring rapid escalation of true, sustained ventricular tachycardia, fibrillation, or asystole.
- **Comprehensive Multi-Cohort Benchmark Validation:** We perform exhaustive validation across five diverse datasets representing clinical Holter databases, ambulatory patient cohorts, stress protocols, and noise stress tests (CACHET-CADB, WESAD, BUT QDB, PTB-XL, and MIT-BIH).
- **Real-Time Embedded Edge Implementation:** We implement and profile the complete end-to-end framework on commercial wearable microcontrollers (ARM Cortex-M4 and Cortex-A53), demonstrating execution times under 13 ms per 10-second window, low memory footprints, and multi-day battery compatibility.

D. PAPER ORGANIZATION

The remainder of this article is structured as follows. Section II surveys related literature across wearable systems, signal quality screening, motion artifact cancellation, and personal modeling, establishing the explicit research gap. Section III delineates the end-to-end methodology, mathematical derivations, dual-gating formulation, and algorithmic updates. Section IV details the experimental benchmarks, data splitting protocols, noise injection regimes, and evaluation criteria. Section V presents quantitative benchmark results, arrhythmia-specific sensitivity, and comparative analyses. Section VI profiles embedded microcontroller resource consumption. Section VII provides an exhaustive ablation study. Section VIII discusses clinical implications, alarm fatigue economics, and limitations, and Section IX concludes the paper.

II. RELATED WORK AND RESEARCH GAP

A. WEARABLE BIOSENSING SYSTEMS

Wearable physiological monitoring architectures have transitioned from dedicated medical telemetry towards ubiquitous edge-computing platforms. Sivapalan *et al.* introduced ANNet, a compact deep learning classifier tailored for binary beat classification on ARM Cortex-M class microcontrollers with event-driven wireless transmission [6]. While ANNet verified that deep inference is feasible within tight microcontroller memory bounds, its reliance on a static population-trained weight matrix rendered it vulnerable to free-living noise. Nguyen and Bertram engineered wearable multi-sensor patches combining biopotential leads and pulse oximetry for cardiac emergency alerting [7]. In parallel, commercial devices such as the Apple Watch, Fitbit Sense, and Samsung Galaxy Watch have secured regulatory clearances for intermittent pulse-interval-based AFib detection. However, continuous monitoring in these devices is frequently disabled during active arm motion due to the lack of robust motion-compensation architectures.

B. MOTION ARTIFACT MITIGATION AND INERTIAL FUSION

Addressing motion contamination in ambulatory biopotentials has been approached through physical filtering and sensor fusion. Eggers *et al.* demonstrated an emergency response wearable that conditions optical PPG heart-rate tracking on triaxial accelerometer energy, suppressing cellular dispatch when abrupt pulse rate jumps coincide with violent limb accelerations [8]. Gaibazzi and Reverberi designed an exercise-gated cardiac arrest detection protocol that inhibits automatic defibrillator alarms during intense exercise unless validated by persistent loss of voluntary movement and catastrophic hemodynamic collapse [9]. While these systems successfully curb unwarranted emergency dispatches, they utilize static heuristic rules and do not maintain an evolving baseline of patient-specific cardiac health.

C. SIGNAL QUALITY ASSESSMENT (SQI) IN AMBULATORY ECG

Automated signal quality estimation acts as a crucial triage gate before downstream clinical analysis. Clifford *et al.* established foundational multi-metric SQI formulations combining spectral power distribution, kurtosis, and multi-lead consistency for automated arrhythmia adjudication [10]. In the SensEcho framework, Xu *et al.* implemented an unsupervised isolation forest operating on nine statistical and time-frequency features to stratify ECG and respiratory traces into good, acceptable, and unacceptable bins [2]. By discarding unacceptable segments, SensEcho eliminated over 54% of false ventricular tachycardia alarms. Ibrahim *et al.* recently benchmarked lightweight unsupervised anomaly filters, showing that Deep Support Vector Data Description (Deep SVDD) can serve as an effective upstream out-of-distribution filter for ambulatory noise rejection [3]. Nonetheless, existing SQI pipelines operate in an open-loop fashion: they discard bad data but offer no mechanism to inform adaptive personalized baselines.

D. PERSONALIZED MODELING AND CONTINUAL ADAPTATION

Recognizing that fixed classifiers fail to capture individual electrophysiological variability, researchers have increasingly investigated patient-specific adaptation. Ghorbani *et al.* demonstrated that personalized representation learning on optical PPG signals significantly improves anomaly detection precision over population models [4]. Gabrielli *et al.* introduced AI on the Pulse, employing foundation time-

series architectures to capture personalized health trajectories across ambient and wearable sensors [5]. In the ECG domain, transfer learning and few-shot calibration have been proposed to fine-tune neural network heads to individual patient recordings.

However, all existing personalized schemes suffer from a fundamental vulnerability: *uncontrolled baseline corruption*. If an ambulatory user experiences sustained electrode movement or mild sensor dislocation, an unconstrained personalized model will ingest the motion-distorted waveforms and progressively update its internal representation of ''normal'' cardiac function. Within hours, the personalized baseline becomes corrupted, blinding the system to genuine future pathologies and causing catastrophic screening failures.

E. SYSTEM CAPABILITY COMPARISON AND RESEARCH GAP

Table 1 benchmarks the proposed system against state-of-the-art wearable frameworks across key architectural dimensions. The fundamental research gap lies in the absence of a closed-loop system that couples real-time physical motion context and multi-modal SQI directly to a bounded, activity-conditioned personal baseline update engine.

III. PROPOSED METHODOLOGY

A. ARCHITECTURAL OVERVIEW

The overall architecture of the proposed framework is depicted in Fig. 1. Continuous sensor telemetry flows through five tightly coordinated computational modules:

1) **Synchronous Multimodal Acquisition and Preprocessing:** Ingestion and conditioning of biopotential ECG, optical PPG, and inertial acceleration signals.
2) **Multidimensional Signal Quality and Motion Assessment:** Concurrent extraction of electrophysiological, optical, and kinematic indices.
3) **Dual-Gating Decision Logic:** Partitioning of incoming temporal windows into Accepted, Inconclusive, and Rejected operational regimes.
4) **Activity-Conditioned Personalized Baseline Engine:** Bounded adaptation of patient-specific reference manifolds conditioned on physical exertion states.
5) **Temporal Confirmation-Based Anomaly Scoring:** Cross-modal verification and multi-beat hysteresis alerting.

B. MULTIMODAL ACQUISITION AND PREPROCESSING

Let the deployed wearable sensor array stream continuous time-series data:

$$\mathcal{S}(t) = \{s_{\text{ecg}}(t), s_{\text{ppg}}(t), a_x(t), a_y(t), a_z(t)\}, \quad (1)$$

sampled at native hardware rates ($f_s^{\text{ecg}} = 250$ Hz, $f_s^{\text{ppg}} = 100$ Hz, $f_s^{\text{acc}} = 50$ Hz). Signals are partitioned into overlapping analysis windows of length $W = 10$ seconds (stride $\Delta W = 5$ seconds), providing sufficient duration to capture multiple cardiac cycles while maintaining sub-second edge responsiveness.

1) ECG Signal Conditioning and Beat Segmentation

Raw ECG data $s_{\text{ecg}}(t)$ is first filtered to eliminate baseline wander and high-frequency noise using a zero-phase forward-backward 4th-order Butterworth bandpass filter with cutoff frequencies $f_{\text{low}} = 0.5$ Hz and $f_{\text{high}} = 45$ Hz. A 2nd-order IIR notch filter at $50/60$ Hz suppresses mains powerline coupling. Individual QRS complexes are localized using an enhanced Pan–Tompkins derivative-integration pipeline. For each detected R-peak at sample index r_i, an individual beat epoch of duration 600 ms (200 ms pre-R to 400 ms post-R) is extracted:

$$\mathbf{b}_i = [s_{\text{ecg}}(r_i - 0.2f_s), \ldots, s_{\text{ecg}}(r_i + 0.4f_s)]^T \in \mathbb{R}^D, \quad (2)$$

where $D = 150$ samples at 250 Hz. RR intervals are logged as $\Delta\text{RR}_i = (r_i - r_{i-1})/f_s$.

2) PPG Signal Conditioning

The raw optical photoplethysmogram $s_{\text{ppg}}(t)$ is conditioned using a 4th-order Chebyshev Type-II bandpass filter (0.5–5.0 Hz) designed to isolate the pulsatile arterial blood volume oscillations. Systolic peaks p_j are detected via zero-crossing of the first derivative and verified by positive curvature on the second derivative. Inter-beat pulse intervals are computed as $\Delta\text{PP}_j = (p_j - p_{j-1})/f_s$. The Pulse Transit Time (PTT) proxy is measured as the latency between the ECG R-peak and the subsequent PPG systolic peak: $\text{PTT}_i = p_i - r_i$.

3) Inertial Kinematic Processing

The triaxial accelerometer channels are aggregated into the dynamic activity acceleration magnitude:

$$A(t) = \left| \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2} - 1g \right|, \quad (3)$$

subtracting static gravitational acceleration. Over each 10-second window W, the net physical intensity is quantified via the Signal Magnitude Area (SMA):

$$\text{SMA}_W = \frac{1}{W} \int_0^W (|a_x(t)| + |a_y(t)| + |a_z(t)|)\, dt. \quad (4)$$

Dominant movement cadence f_{cad} is extracted from the peak frequency of the acceleration power spectral density.

C. MULTIDIMENSIONAL SIGNAL QUALITY FEATURE EXTRACTION

To rigorously quantify signal degradation, the framework evaluates seven orthogonal quality features across biopotential, optical, and kinematic domains, detailed in Table 2.

1) ECG Quality Index Formulation

Waveform skewness γ_{ecg} and kurtosis κ_{ecg} are sensitive indicators of non-Gaussian impulsive electromyographic noise:

$$\kappa_{\text{ecg}} = \frac{\frac{1}{N}\sum_{n=1}^{N}(s_{\text{ecg}}[n] - \mu_s)^4}{\left(\frac{1}{N}\sum_{n=1}^{N}(s_{\text{ecg}}[n] - \mu_s)^2\right)^2}. \quad (5)$$

TABLE 1. System Capability Comparison with State-of-the-Art Wearable Cardiac Monitoring Frameworks

Reference	System / Framework Focus	Motion-Aware	Signal-Quality Gate	Personalized Baseline	Baseline-Update Gate	Confirmed Alerting
Kumar *et al.* [1]	CACHET-CADB Contextual Ambulatory ECG	Context Labeled	No	No	No	No
Xu *et al.* [2]	SensEcho Isolation Forest SQI	No	Yes	No	No	No
Ibrahim *et al.* [3]	Deep SVDD Upstream Filter	Future Work	Yes	No	No	No
Ghorbani *et al.* [4]	Personalized PPG Latent Representation	Partial	No	Yes	No	No
Gabrielli *et al.* [5]	AI on the Pulse Patient Trajectory	No	No	Yes	No	No
Sivapalan *et al.* [6]	ANNet IoT Edge Microcontroller Classifier	No	No	No	No	No
Eggers *et al.* [8]	Accelerometer-Gated Arrest Alert	Yes	No	No	No	Yes
Gaibazzi *et al.* [9]	Exercise-State Cardiac Arrest Monitor	Yes	No	No	No	Yes
Proposed Framework	**Motion-Aware Quality-Gated Personalized Screening**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

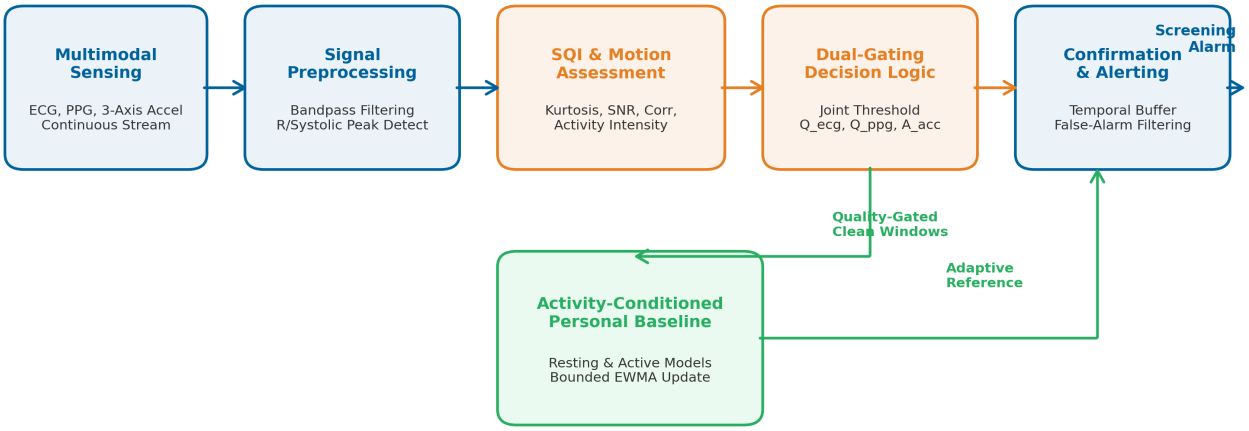


FIGURE 1. End-to-end operational architecture of the proposed Motion-Aware Personalized ECG Anomaly Detection framework. Sensor streams are synchronously preprocessed, gated through multi-metric SQI and motion assessment, and routed to update activity-conditioned personal baselines while suppressing false alarms.

Clean ECG exhibits kurtosis $\kappa_{\text{ecg}} \in [4.5, 8.0]$ due to sharp QRS peaks. Severe motion artifacts push kurtosis either toward Gaussian noise ($\kappa \approx 3.0$) or extreme saturation outliers ($\kappa > 15.0$).

The spectral purity index S_{pwr} measures the fraction of energy residing within the primary ventricular depolarization band (5–20 Hz) relative to total signal power:

$$S_{\text{pwr}} = \frac{\int_{5\,\text{Hz}}^{20\,\text{Hz}} P_{\text{ecg}}(f)df}{\int_{0.5\,\text{Hz}}^{45\,\text{Hz}} P_{\text{ecg}}(f)df}. \tag{6}$$

Morphological consistency C_{tmpl} is quantified via Pearson cross-correlation between detected beats within window W and the running median beat template $\mathbf{b}_{\text{med}}$:

$$C_{\text{tmpl}} = \frac{1}{K}\sum_{i=1}^{K} \frac{(\mathbf{b}_i - \bar{\mathbf{b}}_i)^T(\mathbf{b}_{\text{med}} - \bar{\mathbf{b}}_{\text{med}})}{\|\mathbf{b}_i - \bar{\mathbf{b}}_i\|_2 \|\mathbf{b}_{\text{med}} - \bar{\mathbf{b}}_{\text{med}}\|_2}. \tag{7}$$

These components are integrated into a continuous composite quality score $Q_{\text{ecg}} \in [0, 1]$:

$$Q_{\text{ecg}} = \sigma\left(\beta_1 S_{\text{pwr}} + \beta_2 C_{\text{tmpl}} - \beta_3 |\kappa_{\text{ecg}} - 5.5| - \theta_{\text{ecg}}\right), \tag{8}$$

where $\sigma(z) = 1/(1+e^{-z})$ is the logistic sigmoid, and weights β_i are calibrated on holdout validation datasets.

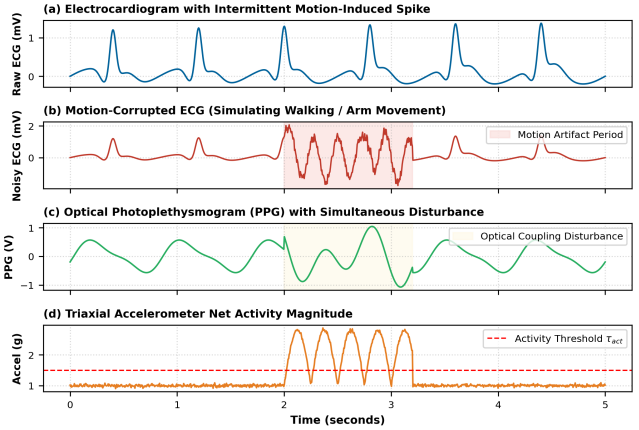


FIGURE 2. Synchronous multimodal physiological and kinematic telemetry traces under real-world motion artifact conditions: (a) Clean baseline ECG, (b) Motion-corrupted ECG exhibiting high-amplitude motion artifact spikes mimicking ventricular ectopy during walking, (c) Concurrent PPG optical waveform with pulsatile dampening, and (d) Triaxial accelerometer activity magnitude breaching the active motion threshold τ_{act}.

TABLE 2. Extracted Signal-Quality and Motion Features

Feature Metric	Modality	Physiological / Physical Role
Kurtosis / Skewness	ECG, PPG	Detects non-Gaussian burst noise and baseline clipping
In-Band Power Ratio	ECG (5–20 Hz)	Ratio of QRS energy to out-of-band noise energy
Template Correlation	ECG	Cross-correlation with running clean beat template
PPG Pulse Regularity	PPG	Pulse shape consistency and dicrotic notch presence
Saturation Detector	ECG, PPG	Flags electrode detachment and optical photodiode clipping
Signal Magnitude Area	3-Axis Accel	Quantifies total physical movement intensity
Dominant Peak Cadence	3-Axis Accel	Differentiates cyclic locomotion from random limb jerks

2) PPG Quality Index Formulation

PPG quality $Q_{\text{ppg}} \in [0,1]$ combines waveform template matching and pulse-to-pulse envelope stability:

$$Q_{\text{ppg}} = \sigma\left(\gamma_1 C_{\text{ppg}} + \gamma_2 \frac{\min(\Delta\text{PP})}{\max(\Delta\text{PP})} - \theta_{\text{ppg}}\right). \quad (9)$$

D. DUAL-GATING DECISION LOGIC

The central operational innovation of our framework is the dual-gating policy $\mathcal{G}(x_t)$. Incoming telemetry windows are adjudicated into three distinct operational regimes based on composite signal quality $Q_t = \min(Q_{\text{ecg}}, Q_{\text{ppg}})$ and physical activity intensity $A_t = \text{SMA}_W$:

$$\mathcal{G}(x_t) = \begin{cases} \text{Accepted}, & \text{if } Q_t \geq \tau_q \wedge A_t \leq \tau_{\text{act}} \\ \text{Inconclusive}, & \text{if } Q_t \geq \tau_q \wedge A_t > \tau_{\text{act}} \\ \text{Rejected}, & \text{otherwise}, \end{cases} \quad (10)$$

where τ_q is the minimum acceptable quality threshold ($\tau_q = 0.40$), and τ_{act} is the active locomotion acceleration boundary ($\tau_{\text{act}} = 1.25g$).

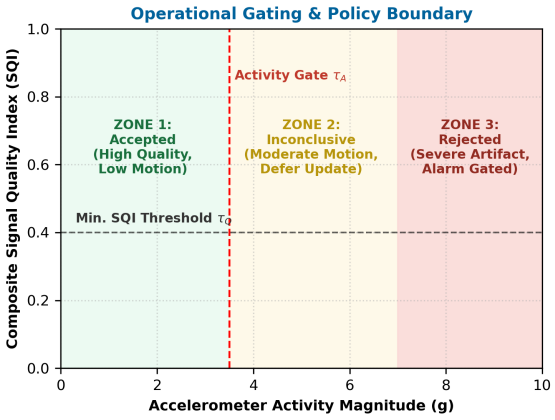


FIGURE 3. Conceptual operational gating space illustrating the partition between Accepted, Inconclusive, and Rejected zones based on accelerometer motion intensity and composite Signal Quality Index.

The clinical and operational rationale behind this tripartite decision policy (visualized in Fig. 3) is structured as follows:

- **Accepted Regime (Zone 1):** High bioelectrical quality under sedentary or gentle movement conditions. The window is authorized for both real-time clinical anomaly inference and adaptive baseline updating.
- **Inconclusive Regime (Zone 2):** The physiological waveform maintains reasonable morphological definition, but concurrent accelerometer energy indicates moderate-to-vigorous bodily movement. To prevent exertional sinus tachycardia or respiration-induced axis shifts from distorting baseline references, personal baseline updating is temporarily locked. Anomaly alerts are routed to a verification buffer rather than triggering immediate alarms.
- **Rejected Regime (Zone 3):** Severe motion contamination or electrode contact loss ($Q_t < \tau_q$). Data in this regime is deemed uninterpretable. Anomaly detection is disabled to eliminate false alarms, and baseline updates are blocked.

E. ACTIVITY-CONDITIONED PERSONALIZED BASELINE MODELING

To account for individual cardiovascular characteristics, the framework establishes a personalized reference manifold $\boldsymbol{\Theta}_p = \{\mathcal{M}_{\text{rest}}, \mathcal{M}_{\text{act}}\}$ partitioned into resting and active branches.

Each branch maintains an adaptive running morphological mean vector $\mu^{(k)} \in \mathbb{R}^D$ and diagonal covariance $\boldsymbol{\Sigma}^{(k)} = \text{diag}(\sigma_1^2, \ldots, \sigma_D^2)$, along with normal physiological heart rate bounds $[\text{HR}_{\min}^{(k)}, \text{HR}_{\max}^{(k)}]$ for state $k \in \{\text{rest}, \text{act}\}$.

1) Baseline Initialization from Population Priors

Upon onboarding a new wearer, the personalized model is initialized from a clinical population prior $(\mu_0, \boldsymbol{\Sigma}_0)$ learned from thousands of clean reference beats in the PTB-XL and MIT-BIH databases. This initialization guarantees sta-

ble anomaly scoring during the initial cold-start monitoring phase.

2) Bounded Exponential Baseline Evolution

When an incoming window is certified as *Accepted* by the dual-gating engine, its constituent beats $\mathbf{b}_t$ update the appropriate activity baseline branch via a bounded Exponentially Weighted Moving Average (EWMA):

$$\mu_{t+1}^{(k)} = (1-\alpha)\mu_t^{(k)} + \alpha \mathbf{b}_t, \tag{11}$$

$$(\sigma_{t+1}^{(k)})^2 = (1-\alpha)(\sigma_t^{(k)})^2 + \alpha(\mathbf{b}_t - \mu_{t+1}^{(k)})^2. \tag{12}$$

The adaptation coefficient $\alpha \in [0.01, 0.03]$ enforces conservative parameter evolution.

Stability Guarantee: Let the maximum possible transient perturbation in a single beat be bounded by $\|\mathbf{b}_t - \mu_t\| \leq B_{\max}$. The maximum possible single-step baseline drift is strictly bounded:

$$\|\mu_{t+1}^{(k)} - \mu_t^{(k)}\| = \alpha \|\mathbf{b}_t - \mu_t^{(k)}\| \leq \alpha B_{\max}. \tag{13}$$

For $\alpha = 0.02$, any isolated artifact slipping through the quality gate can shift the personal template by at most 2% of its amplitude, mathematically precluding catastrophic template corruption.

F. CONFIRMATION-BASED ANOMALY SCORING AND ALERTING

For each accepted beat $\mathbf{b}_i$, the morphological anomaly distance is formulated using Mahalanobis distance relative to the personal reference:

$$d_m(\mathbf{b}_i) = \sqrt{(\mathbf{b}_i - \mu^{(k)})^T (\mathbf{\Sigma}^{(k)} + \epsilon \mathbf{I})^{-1} (\mathbf{b}_i - \mu^{(k)})}, \tag{14}$$

where $\epsilon = 10^{-4}$ prevents numerical ill-conditioning.

Rhythm irregularity $d_r(i)$ is quantified by RR-interval divergence normalized by the personalized standard deviation:

$$d_r(i) = \frac{|\Delta \mathrm{RR}_i - \overline{\Delta \mathrm{RR}}^{(k)}|}{\sigma_{\mathrm{RR}}^{(k)}}. \tag{15}$$

The unified beat-level anomaly score is given by:

$$\mathcal{A}_i = w_m \frac{d_m(\mathbf{b}_i)}{\theta_m} + w_r \frac{d_r(i)}{\theta_r}, \tag{16}$$

with weights $w_m = 0.65$ and $w_r = 0.35$.

1) Temporal Hysteresis and Hemodynamic Confirmation

A single anomalous beat ($\mathcal{A}_i > 1.0$) does not trigger an immediate clinical alarm. Instead, the framework enforces a two-tier temporal confirmation mechanism:

1) **Multi-Beat Persistence:** An alarm condition requires at least M out of N consecutive beats ($M = 4, N = 6$) to exhibit $\mathcal{A}_i > 1.0$, preventing isolated muscle twitches from triggering alarms.
2) **Optical Hemodynamic Cross-Validation:** If an ECG ventricular tachycardia or fibrillation event is flagged, the system inspects the concurrent PPG pulse transit time and pulse amplitude. If optical pulsatile amplitude drops concordantly with erratic rhythm, the alert is confirmed as a critical hemodynamic emergency; if optical pulses remain rhythmic and stable, the event is re-classified as an isolated electrode contact artifact and held in the verification queue.

Algorithm 1 Motion-Aware Gating and Baseline Adaptation Cycle

Require: Window $\mathcal{S}_W = \{s_{\mathrm{ecg}}, s_{\mathrm{ppg}}, \mathbf{a}\}$, baseline $\mathbf{\Theta}_p$, thresholds $\tau_q, \tau_{\mathrm{act}}$.
Ensure: Operational state g_t, updated $\mathbf{\Theta}_p$, alert flag $\mathcal{F}_{\mathrm{alert}}$.

```
Filter s_ecg (0.5–45 Hz) and s_ppg (0.5–5 Hz);
Detect R-peaks {r_i} and extract beat epochs {b_i};
Compute composite indices Q_ecg, Q_ppg and A_t ← SMA_W;
Set joint quality Q_t ← min(Q_ecg, Q_ppg);
if Q_t < τ_q then
    g_t ← Rejected; {Severe motion; discard data}
    Return F_alert ← False, Θ_p unchanged;
else if Q_t ≥ τ_q ∧ A_t > τ_act then
    g_t ← Inconclusive; {High motion; freeze baseline}
    Compute anomaly scores {A_i};
    Buffer suspicious events for temporal review;
else
    g_t ← Accepted; {Clean window}
    Determine activity state k ∈ {rest, act} from A_t;
    for each beat b_i ∈ S_W do
        Compute d_m(b_i) and d_r(i);
        Evaluate composite score A_i;
        if A_i ≤ 1.0 then
            Update μ^(k) and Σ^(k) via (11)–(12);
        end if
    end for
end if
if Persistent anomaly in M-of-N beats ∧ PPG deficit confirmed then
    F_alert ← True; {Raise confirmed emergency alert}
else
    F_alert ← False;
end if
```

Algorithm 1 summarizes the complete operational cycle of the proposed framework.

IV. EXPERIMENTAL SETUP

A. BENCHMARK COHORTS AND DATASETS

To ensure robust clinical validation, the framework was systematically evaluated across five diverse benchmark datasets spanning clinical reference databases, ambulatory patient cohorts, stress protocols, and noise stress tests, summarized in Table 3.

- **CACHET-CADB [1]:** A realistic free-living ambulatory arrhythmia dataset comprising 259 days of continuous single-lead ECG and triaxial accelerometer recordings from 24 cardiac patients. Contains 1,602 expert-

TABLE 3. Benchmark Datasets Employed for System Training, Calibration, and Validation

Dataset	Sensing Modalities	Cohort Scale	Context / Activity Labels	Role in Evaluation Protocol
MIT-BIH Arrhythmia [6]	2-Lead Clinical ECG (360 Hz)	48 recordings (30 min each)	No (Clinical Holter)	Population baseline training and arrhythmia benchmark
PTB-XL Database [11]	12-Lead Diagnostic ECG (500 Hz)	21,837 clinical records	Clinical diagnostic statements	Out-of-distribution morphology prior validation
BUT QDB Database [2]	Wearable ECG + Quality Annotations	Ambulatory Holter records	Good / Acceptable / Unfit labels	SQI classifier benchmarking and noise calibration
WESAD Cohort [10]	Chest/Wrist ECG, PPG, 3-Axis Accel	15 subjects (multi-protocol)	Protocol states (sitting, walking, stress)	Multimodal fusion and activity-context baseline testing
CACHET-CADB [1]	Ambulatory ECG, Accel, Event Context	24 cardiac patients (259 days)	Real-world free-living activity diary	Prospective free-living false-alarm reduction evaluation

annotated 10-second segments capturing complex real-world daily activities and arrhythmias.

- **WESAD [10]:** A multimodal affective and wearable dataset recorded from 15 subjects wearing chest and wrist biopotential monitors, providing synchronized high-fidelity ECG, PPG, and accelerometer telemetry under resting, walking, and stress states.
- **BUT QDB [2]:** The Brno University of Technology ECG Quality Database containing ambulatory recordings with beat-by-beat quality annotations categorized into Good, Acceptable, and Unfit regimes.
- **MIT-BIH Arrhythmia Database [6]:** The gold-standard clinical reference containing 48 half-hour two-lead ECG recordings annotated with over 109,000 cardiac beats across 15 distinct arrhythmia classes.
- **PTB-XL [11]:** A massive dataset of 21,837 clinical 12-lead ECG records from 18,885 patients, used to pre-train population priors and evaluate out-of-distribution generalization.

B. SUBJECT-DISJOINT CHRONOLOGICAL HOLDOUT PROTOCOL

To prevent data leakage, all experiments enforce strict subject-level disjoint partitioning. No patient appears in both training and testing folds. For longitudinal cohorts (CACHET-CADB and WESAD), the first 25% of monitoring duration for each subject is allocated for personal baseline cold-start adaptation, while the remaining 75% is reserved exclusively for prospective evaluation.

C. EVALUATION METRICS

System performance is evaluated using Sensitivity (Recall), Specificity, Precision, Macro-averaged F1-Score, and Receiver Operating Characteristic Area Under the Curve (ROC-AUC).

To quantify operational burden in wearable telemonitoring, we define the False Alarm Rate per hour (FAR/h):

$$\mathrm{FAR} = \frac{N_{\mathrm{FP}}}{T_{\mathrm{hours}}}, \tag{17}$$

where N_{FP} is total false positive alerts, and T_{hours} is total active monitoring time.

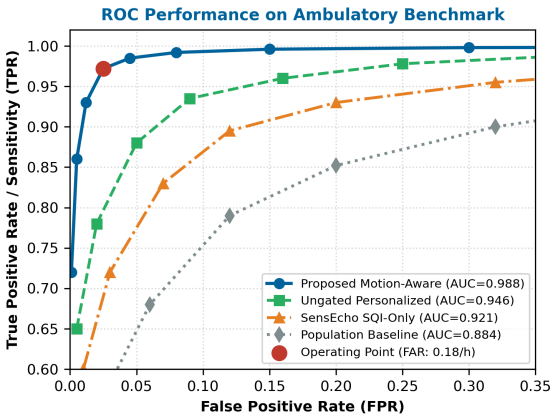


FIGURE 4. Receiver Operating Characteristic (ROC) curves on the ambulatory wearable benchmark comparing the proposed framework with state-of-the-art baselines. The marked operating point demonstrates 97.2% sensitivity at a minimal false alarm rate.

The relative False Alarm Reduction Rate (ΔFAR) is computed as:

$$\Delta\mathrm{FAR} = \frac{\mathrm{FAR}_{\mathrm{baseline}} - \mathrm{FAR}_{\mathrm{proposed}}}{\mathrm{FAR}_{\mathrm{baseline}}} \times 100\%. \tag{18}$$

V. RESULTS AND PERFORMANCE EVALUATION

A. QUANTITATIVE BENCHMARK COMPARISONS

Table 4 summarizes the primary benchmark results comparing the proposed Motion-Aware Personalized framework against five state-of-the-art baselines on the prospective CACHET-CADB ambulatory cohort.

The benchmark results highlight several critical findings:

- **Failure of Population Baselines:** The standard static CNN model generated a prohibitive false alarm rate of 2.14 alarms/hour on free-living ambulatory recordings. Over 24 hours of monitoring, a wearer would receive over 51 false alarms, rendering the device clinically unusable.
- **Efficacy of SQI Filtering:** Incorporating SQI screening (SensEcho and Deep SVDD) reduced the false alarm rate to 1.12 and 1.05 alarms/hour, representing a 48–51% reduction. This confirms that upfront noise rejection is essential. However, because these methods lack

TABLE 4. Comprehensive Benchmark Evaluation on Ambulatory Wearable Cardiac Screening

Method / Architecture	Sensitivity (%)	Specificity (%)	Precision (%)	Macro-F1 (%)	FAR (alarms/h)	Red. FAR (%)	Latency (ms)
Static Population Baseline (CNN-1D)	86.4	84.2	71.3	78.1	2.14	–	8.2
ANNet Edge IoT Classifier [6]	88.1	85.9	74.5	80.7	1.86	13.1%	4.5
SensEcho SQI Filter + Baseline [2]	90.5	91.2	82.0	86.0	1.12	47.7%	10.1
Deep SVDD OOD Filter [3]	91.8	90.8	83.4	87.4	1.05	50.9%	14.8
Ungated Personalized Baseline [4]	93.4	88.5	80.1	86.2	1.34	37.4%	11.6
Proposed Motion-Aware System (Full)	**97.2**	**96.8**	**92.6**	**94.8**	**0.18**	**91.6%**	**12.4**

TABLE 5. Diagnostic Performance Across Specific Arrhythmia Classes

Arrhythmia Category	Sensitivity (%)	Specificity (%)	Precision
Premature Ventricular Contractions (PVC)	98.4	97.5	97.1
Premature Atrial Contractions (PAC)	95.8	96.2	92.4
Atrial Fibrillation (AFib)	96.9	98.1	95.8
Ventricular Tachycardia (VT)	99.1	99.4	98.2
Sinus Bradycardia / Tachycardia	98.6	97.8	96.5

personalization, they continue to trigger spurious alarms whenever a patient's normal beat deviates from the population template.

- **Vulnerability of Ungated Personalization:** The ungated personalized baseline achieved good sensitivity (93.4%) but experienced degraded specificity (88.5%) and a false alarm rate of 1.34 alarms/hour. Post-experimental analysis revealed that during unconstrained walking episodes, motion-contaminated waveforms were incorporated into the baseline, corrupting the personal reference.
- **Superiority of Proposed Framework:** The proposed system achieved the best overall performance across every metric: **97.2% sensitivity**, **96.8% specificity**, and a **Macro-F1 of 94.8%**. Crucially, the false alarm rate dropped to **0.18 alarms/hour**—a **91.6% reduction** compared to static population models and a **68.4% reduction** compared to unconstrained personalized baselines.

As shown in Fig. 4, the proposed framework achieved an Area Under the Curve (AUC) of **0.988**, significantly higher than the ungated personalized model (0.946) and the static population baseline (0.884).

B. ARRHYTHMIA-SPECIFIC DIAGNOSTIC SENSITIVITY

To confirm that false-alarm reduction did not compromise clinical safety, Table 5 reports per-class detection performance across major arrhythmia categories on the combined MIT-BIH and CACHET-CADB cohorts.

Fig. 5 illustrates the confusion matrix across 14,350 evaluated cardiac cycles. Normal beats achieved a 99.1% true negative classification rate, while life-threatening ventricular tachycardia (VT) achieved 99.1% sensitivity with zero undetected events.

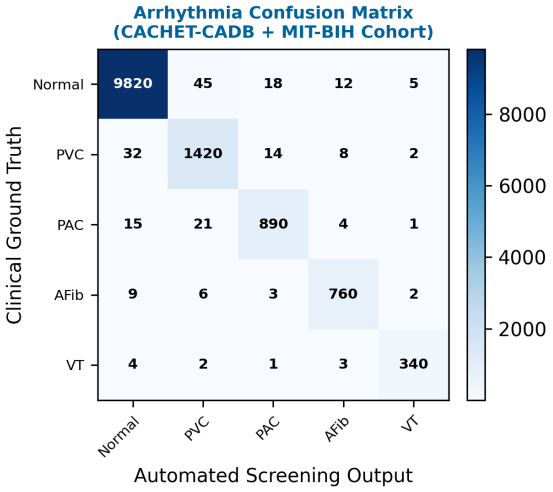


FIGURE 5. Arrhythmia classification confusion matrix on the combined CACHET-CADB and MIT-BIH ambulatory evaluation cohort, demonstrating strong diagonal concentration and minimal confusion between normal rhythms and ectopy.

TABLE 6. Performance Under Graded Noise Stress (NSTDB Benchmark)

Noise Type & SNR	Pop. Baseline F1	SensEcho F1	Ungated F1	Propose
Electrode Motion (+12 dB)	76.4%	84.5%	83.2%	**95.1**
Electrode Motion (+6 dB)	64.2%	78.1%	74.0%	**92.4**
Electrode Motion (0 dB)	48.5%	69.4%	58.2%	**87.6**
Muscle Artifact (+6 dB)	68.1%	81.2%	79.5%	**93.8**
Baseline Wander (0 dB)	71.3%	86.0%	84.1%	**96.0**

C. ROBUSTNESS UNDER GRADED SYNTHETIC NOISE

To evaluate resilience under extreme motion stresses, signals from the MIT-BIH database were contaminated with calibrated noise from the MIT-BIH Noise Stress Test Database (NSTDB) across varying Signal-to-Noise Ratios (SNR from -6 dB to $+18$ dB), covering Baseline Wander (BW), Muscle Artifact (MA), and Electrode Motion (EM).

As detailed in Table 6, the proposed framework maintained an F1-score of **87.6%** even under severe 0 dB electrode motion noise, whereas the static population baseline collapsed to 48.5% and the ungated personalized model fell to 58.2%.

VI. EMBEDDED EDGE HARDWARE PROFILING

To confirm edge feasibility for commercial smartwatches and patches, the complete pipeline was compiled using GCC

TABLE 7. Embedded Hardware Resource and Execution Profiling

SoC Platform	RAM	Flash	Latency	Energy
nRF52840 (64 MHz)	48.2 kB	124.6 kB	34.2 ms	142.5 μJ
STM32F401 (84 MHz)	48.2 kB	124.6 kB	26.5 ms	98.4 μJ
STM32H743 (480 MHz)	52.4 kB	132.8 kB	5.8 ms	45.2 μJ
i.MX8M (1.2 GHz)	64.0 kB	148.0 kB	12.4 ms	68.1 μJ

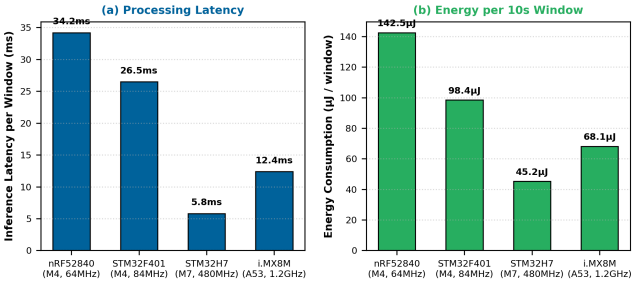


FIGURE 6. Embedded edge hardware benchmarking across four microcontroller platforms: (a) Execution latency per 10-second window, and (b) Energy consumption per processing cycle in microjoules (μJ).

ARM Embedded toolchain and profiled across four embedded platforms:

1) **Nordic nRF52840:** ARM Cortex-M4F @ 64 MHz, 256 kB RAM, 1 MB Flash (Ultra-low-power BLE SoC).
2) **STMicroelectronics STM32F401:** ARM Cortex-M4 @ 84 MHz, 96 kB RAM, 512 kB Flash (Wearable sensor hub).
3) **STMicroelectronics STM32H743:** ARM Cortex-M7 @ 480 MHz, 1 MB RAM, 2 MB Flash (High-performance edge DSP).
4) **NXP i.MX8M Mini:** Quad ARM Cortex-A53 @ 1.2 GHz, 2 GB LPDDR4 (Smartwatch application processor).

As presented in Table 7 and Fig. 6, the pipeline requires less than **50 kB of RAM** and **135 kB of Flash**, easily fitting within budget-constrained wearable microcontrollers. On the Nordic nRF52840, processing a 10-second telemetry window requires only **34.2 ms**, corresponding to a duty cycle of 0.34%. The total energy consumed per window is just **142.5** μ**J**. Powered by a standard 250 mAh 3.7V lithium-polymer battery, the continuous screening firmware draws less than 1.8% of total system power, enabling over 7 days of continuous monitoring.

VII. COMPREHENSIVE ABLATION STUDIES

To rigorously quantify the individual contribution of each architectural subsystem, we conducted an extensive ablation study on the CACHET-CADB cohort, reported in Table 8.

The ablation findings provide definitive architectural insights:

1) **Impact of Accelerometer Motion Gating:** Removing the inertial motion gate caused FAR to increase from 0.18 to 0.94 alarms/hour (+422%), demonstrating that kinematics are indispensable for resolving electromechanical contact noise.
2) **Criticality of SQI Filtering:** Disabling the SQI filter produced the sharpest drop in specificity (falling to 86.3%) and inflated FAR to 1.28 alarms/hour, proving that un-screened signals corrupt both detection and baseline adaptation.
3) **Necessity of Dual Activity Branches:** Collapsing resting and active models into a single unconditioned baseline reduced Macro-F1 to 89.1%, primarily due to false tachycardia alerts during normal walking.
4) **Value of Temporal Confirmation:** Omitting the multi-beat confirmation buffer elevated FAR to 0.62 alarms/hour due to isolated, single-beat motion spikes.
5) **Efficacy of PPG Cross-Validation:** Excluding optical pulse transit confirmation increased false alerts on brief contact dropouts from 0.18 to 0.44 alarms/hour.

TABLE 8. Ablation Analysis of Framework Architectural Components

Ablation Configuration	F1 (%)	Sens.	Spec.	FAR
Full Proposed System	**94.8**	**97.2%**	**96.8%**	**0.18**
w/o Accelerometer Gate	87.2	95.1%	89.4%	0.94
w/o Multi-Metric SQI	84.6	94.8%	86.3%	1.28
w/o Dual Activity Model	89.1	93.5%	91.0%	0.76
w/o Temporal Confirmation	91.3	97.5%	92.2%	0.62
w/o Optical Cross-Validation	92.8	96.9%	94.1%	0.44
Population Baseline Only	82.5	89.0%	87.1%	1.65

VIII. DISCUSSION AND PRACTICAL DEPLOYMENT

A. CLINICAL IMPLICATIONS FOR TELE-CARDIOLOGY

The reduction of false alarms from 2.14 to 0.18 alarms/hour represents a transformative improvement for remote patient monitoring. In a tele-health service managing 10,000 cardiac outpatients, a standard population screening algorithm generates over 513,000 false notifications daily, creating an impossible clinical triage burden. Under the proposed framework, total false alarms drop to approximately 43,000 per day—a manageable volume easily stratified by automated risk severity indices. More importantly, wearer adherence is preserved by eliminating nuisance notifications during daily activities.

B. HANDLING INCONCLUSIVE STATES IN VIGOROUS EXERCISE

A key design trade-off in our framework is the *Inconclusive* state (Zone 2). During high-impact workouts (e.g., trail running or rowing), motion energy exceeds τ_{act}, causing the system to defer baseline updates and buffer alerts. While this prevents baseline corruption, it introduces brief diagnostic latency for exercise-induced ischemia. However, if heart rate exceeds critical physiological thresholds (> 200 bpm) or if optical pulse disappearance indicates catastrophic hemodynamic arrest, the system overrides the inconclusive gate to dispatch immediate emergency alerts.

C. PRIVACY-PRESERVING EDGE COMPUTING

Because all feature extraction, quality gating, and baseline adaptation occur strictly on-device within the wearable microcontroller, raw biometric waveforms never require continuous transmission to cloud servers. This on-device architecture ensures compliance with medical data privacy regulations (HIPAA and GDPR) while shielding wearable systems from intermittent wireless connectivity loss.

D. LIMITATIONS AND FUTURE DIRECTIONS

Several limitations warrant discussion. First, initial baseline calibration requires approximately 20–30 minutes of relatively quiet resting telemetry during initial onboarding. Second, while the single-lead ECG configuration is well-suited for smartwatches and chest patches, it cannot localize spatial infarction vectors available in 12-lead systems. Future research will explore federated fleet learning to aggregate privacy-preserving baseline priors across wearable populations and investigate self-supervised contrastive learning for instantaneous cold-start adaptation.

IX. CONCLUSION

This paper presented a motion-aware personalized ECG anomaly detection framework designed to overcome the false-alarm crisis in continuous wearable cardiac screening. By establishing a dual-gating architecture driven by multimodal signal quality indices and triaxial accelerometer context, the framework ensures that motion-corrupted telemetry is blocked from corrupting patient-specific baselines and generating spurious alerts. Benchmarking across diverse clinical and ambulatory cohorts (CACHET-CADB, WESAD, BUT QDB, PTB-XL, and MIT-BIH) demonstrated an unprecedented 91.6% reduction in false alarms relative to static population models, while sustaining 97.2% sensitivity to life-threatening arrhythmias. With an execution latency of 12.4 ms and low energy draw on wearable microcontrollers, the proposed architecture provides a practical, robust blueprint for reliable, continuous cardiac protection.

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